

JENIL MAKWANA

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Education

Carnegie Mellon University, School of Computer Science

Aug 2026 – Dec 2027 (expected)

Master of Software Engineering

Pittsburgh, PA

- Coursework: Product Management, Decision Statistics, Formal Methods, Embedded Systems, Computer Systems

University of Wisconsin-Madison

Sep 2021 – Dec 2025

Bachelor of Science in Electrical Engineering & Computer Science | GPA: 3.5/4.0

Madison, WI

- Selected Coursework: Algorithms, Database Management Systems, Building User Interfaces, Mathematical Statistics

Skills

Product: Customer discovery, user requirements, product iteration, product launch, cross-functional delivery

Business & Data: Pricing & packaging, survey analysis, adoption metrics, Python, SQL

AI Product: AI product design, LLM integration, document extraction, reliability workflows, human-in-the-loop review

Technical: TypeScript, Next.js, NestJS, Prisma, PostgreSQL, Stripe, C/C++

Experience

Boxmate

May 2024 – Present

Founder

Madison, WI

- Defined [Boxmate](#)'s two-sided marketplace model, registered the LLC and launched to 150+ UW-Madison students, signing 95 hosts and earning \$20K+ in lifetime revenue; worked with lawyers on its terms and conditions.
- Hired and managed an India-based development team and led a cross-functional team of about 10 to ship the Next.js storefront, NestJS/Prisma backend and host mobile app with Stripe checkout and automated host reminders.
- Analyzed 230+ survey responses and customer interviews, finding price sensitivity, low freshman awareness and unmet delivery demand, then set \$29/\$39/\$49 tiers where 88% of 95 paid orders chose Gold, the pickup-and-delivery tier.
- Iterated pricing from per-item to box bundles to flat tiers; summer 2026 revenue grew 3.5x YoY to \$15.3K.
- Designed Boxmate's customer flow on iPad and directed its translation into the UI/UX; worked with a designer on the visual identity and launched social media and UGC creator campaigns through SideShift for summer 2026.
- Gave technical and business pitches at Transcend UW, Fetch Rewards and gener8tor gALPHA, winning \$17.5K.

USEMCO

Mar 2026 – Jul 2026

Electrical Engineer

Tomah, WI

- Identified a manual cutsheet bottleneck and delivered a Python, Next.js and Claude API product adopted by all six engineers across about 18 jobs per week, saving 10–15 minutes per job and flagging missing datasheets.
- Partnered with the EPLAN team on X/Y exports that set datasheet order.
- Designed a two-pass extraction and verification workflow with engineer accept/reject review, feeding corrections back into the AI as electrical rules change; drove adoption of an annual Claude subscription at USEMCO.
- Expanded the tool into searchable job history for reusing schematic sections and tracking cutsheet completion.
- Scoped control-panel orders with sales, resolved client change requests through submittal and resubmittal, and cleared compliance and layout review before releasing EPLAN schematics to the shop floor for assembly.

IdeaFund Ventures

Feb 2026 – Mar 2026

Software Consultant

Madison, WI

- Automated deal-flow intake so website pitch submissions populate Monday.com and notify a partner.
- Supported the firm's adoption of Claude for daily workflows.

Credo Product Development

Jun 2024 – Dec 2025

Software Engineer Intern

McFarland, WI

- Implemented the hardware watchdog and wrote and executed I2C test procedures for an STM32 leak-detection product shipped to a customer, covering sensor bring-up, fault recovery and bench validation in C.
- Built an internal ESP32 irrigation prototype with moisture sensing, valve control, BLE and a mobile app.
- Wrote bare-metal C firmware for an ATtiny3217 pipe-bender motor controller with a DRV8701 H-bridge, 50 kHz PWM, an RTC-backed 1 ms timer engine with interrupt and polled lists, and 50 ms button debouncing.

UW Space Science & Engineering Center

May 2023 – Jun 2024

Software Engineer Intern

Madison, WI

- Replaced manual resistance testing with a configurable Python data pipeline driving a Keithley DAQ6510 and 7700 multiplexer over SCPI/TCP, converting thermistor readings via Steinhart-Hart and logging to HDF5.
- Designed and SPICE-validated a precision [heater-amplifier PCB](#) in Altium for the SHIS instrument and prototyped a Zynq FPGA sensor interface whose bare-metal C++ driver halved prototype harness wiring.

Technical Project

FreeRTOS Multisensor Racing Simulator ([GitHub](#)) | PSoC6, FreeRTOS, Embedded C

Fall 2025

- Owned game-logic and LCD display work for a team-built simulator, coordinating queue-driven input and display tasks at 20 FPS; wrote the RA8875 driver with double buffering and semaphore-protected SPI.